

Power BI Data Modeling & DAX Project Report

Project Title

Data Modeler – Star Schema & DAX Analysis

Schema Design

This project follows a Star Schema where Sales_Fact is the central fact table. Dimension tables include Customer_Dim, Product_Dim, Region_Dim, and Date_Dim. Returns_Fact is treated as a secondary fact table.

Relationships & Filter Flow

All relationships are Many-to-One (*:1) with single direction filtering. The relationship between Returns_Fact and Date_Dim is inactive to avoid ambiguity and is used only when required.

Calculated Columns

- Profit = Revenue - Cost
- ReturnFlag = Returned / Not Returned
- Customer Full Name = FirstName + LastName

Measures

- Total Sales
- Total Cost
- Total Profit
- Return Rate (%)
- Average Sale per Transaction

Time Intelligence

- Year-to-Date (YTD) using TOTALYTD()
- Previous Year using SAMEPERIODLASTYEAR()

Conclusion

The data model is optimized using a Star Schema, proper relationships, and efficient DAX calculations ensuring performance and scalability.